VSN-LSTM Multi-Asset Return Predictor

S26 IC QUANTITATIVE SECTOR MENTORSHIP | SEBIN AHN

PROJECT OVERVIEW

Predicting **t+1 daily log returns** for SPY, TLT, GLD, and XLE using a hybrid VSN-LSTM architecture.

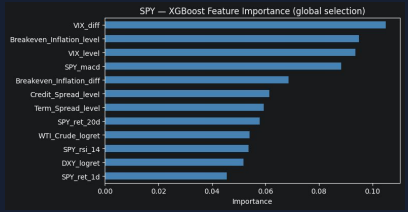
DATA ENGINEERING

- **Refinitiv API:** Daily OHLCV for assets & DXY (Dollar Index).
- **FRED Macro:** Yield spreads, VIX, Breakeven Inflation.
- **Causal Scaling:** Expanding window to eliminate bias.

FEATURE SELECTION

Stage 1: Redundancy Filter
Spearman clustering (threshold 0.90) to de-correlate features.

Stage 2: Relevance Filter
XGBoost ranking based on information gain; Top-12 retained.



MODEL ARCHITECTURE: VSN + LSTM

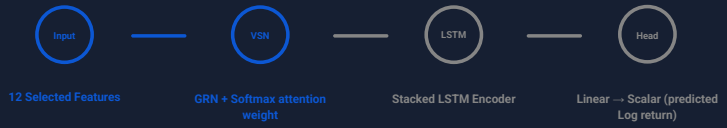
The framework integrates local feature selection with global temporal modeling to handle noisy financial data.

1. Variable Selection Network (VSN)

Uses Gated Residual Networks (GRN) to independently project features and produce softmax attention weights.

2. Stacked LSTM Encoder

Processes sequences of 20 trading days to learn non-linear temporal dependencies. Optimized using Huber Loss.



TRAINING & VALIDATION PROTOCOL

Walk-Forward Validation: Expanding window with 20-day Embargo to prevent serial correlation leakage.

Optuna Tuning: 30 trials using TPE sampler targeting Huber Loss (outlier robust)) on Fold-0 validation set.

Features: 6 Technical indicators per asset + 11 global Macro drivers.

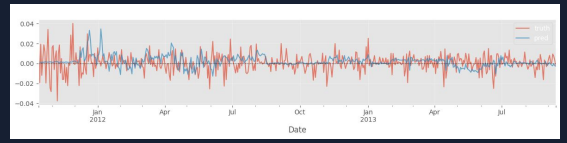


Figure 2: Predicted vs Actual for SPY

RESULTS & METRICS

Model Performance (RMSE)

SPY (Equity):	0.0107
GLD (Gold):	0.0104
TLT (Bonds):	0.0094
XLE (Energy):	0.0171

Asset Performance (Hit Rate)

SPY:	53.3%
GLD:	52.2%
TLT:	51.4%
XLE:	50.4%

CONCLUSION

- Model hit ratios exceed the 50% benchmark, validating predictive edge despite high market noise.
- VSN layer provides clear interpretability by isolating key macro drivers like VIX and Credit Spreads.
- Framework establishes a modular, leak-free DL pipeline for rapid cross-asset strategy testing.
- Future work: Integrate position-sizing directly into loss functions to optimize risk-adjusted returns for larger scale of portfolio.